ecbench report: ECBS1hec2206c6baf4

Label: index calculus (prime-abscissa 32, MITM) against rho over 16..26-bit prime-order curves

Description: Method × size scaling study (prime curves, bits of p 16,18,20,22,24,26; 2 independent single-target workload(s) per curve, 3 interleaved round(s)). One curve per size so `ecbench bound fit` and `ecbench report` can fit cost against r. Counted group operations; a measurement, not a performance claim.

Generated by `ecbench report` (ecbench.report/v1). Observations only; derived numbers carry the key of their derivation (last section).

Session

field	value
session id	ECBS1hec2206c6baf4
spec id	ECS1hbf3d3eb3685f
commit	f793f1fe5edc540668ed836d5e975eb040a34dff (dirty)
status	complete
run options	cpus="auto", lock=/tmp/crypto-bench.lock, wait_for_lock=false, allow_busy=true

Host and isolation

Intel(R) Xeon(R) Processor @ 2.10GHz; 4 logical / 4 physical cores; kernel 6.18.44-fc-v139; 15.7 GiB; env class ECBENV2he1389885ccd0; virtualization cpu_flag=hypervisor, systemd_detect_virt_vm=kvm.

Isolation earned by the 144 measured records: L0 0, L1 11, L2 133, L3 0. Required of a wall-clock figure: L2.

blocker	measured records
L3: governor is None, not performance	144
L3: not known to be bare metal	144
L3: run CPU is in neither isolcpus nor nohz_full	144
L3: turbo/boost is not known to be off	144
L2: 1 ticks of hypervisor steal on the run CPU	6
L2: 54.7 preemptions per second	2
L2: 2 ticks of hypervisor steal on the run CPU	1
L2: 67.5 preemptions per second	1
L2: run-queue delay 0.663% of the solve	1

Arms

arm	role	method	params	expected S [d4]
rho-negation	reference	rho.negation	cap_multiple=64	0.8862
rho-frozen_reference	baseline	rho.frozen_reference	cap_multiple=64	1.2533
ic-pipeline	candidate	ic.pipeline	factor_base=prime-abscissa:size=32; linalg=incremental-gauss; max_trials=100000000; oracle=mitm-prime-batch:m=2; solver=; solver_budget_seconds=0; targets=walk	–
rho-negation-aa	control	rho.negation	cap_multiple=64	0.8862

Workloads

workload	curve	family	log2 r	A	target index	target seed
W43a0abf0d808	icv1-fp16-t295-8d3c3165	prime	15.697	2	0	20261002
Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	prime	15.697	2	1	20261002
Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	prime	17.697	2	0	20261002
Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	prime	17.697	2	1	20261002
W141767324d6e	icv1-fp20-t727-cd198a38	prime	19.101	2	0	20261002
W8d7a772f8ccd	icv1-fp20-t727-cd198a38	prime	19.101	2	1	20261002
W9a300e433cac	icv1-fp22-tm1385-475dcb5f	prime	21.216	2	0	20261002
W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	prime	21.216	2	1	20261002
Wec44ca6ca670	icv1-fp24-t1059-6df599df	prime	23.974	2	0	20261002
W3256bf1f8f74	icv1-fp24-t1059-6df599df	prime	23.974	2	1	20261002

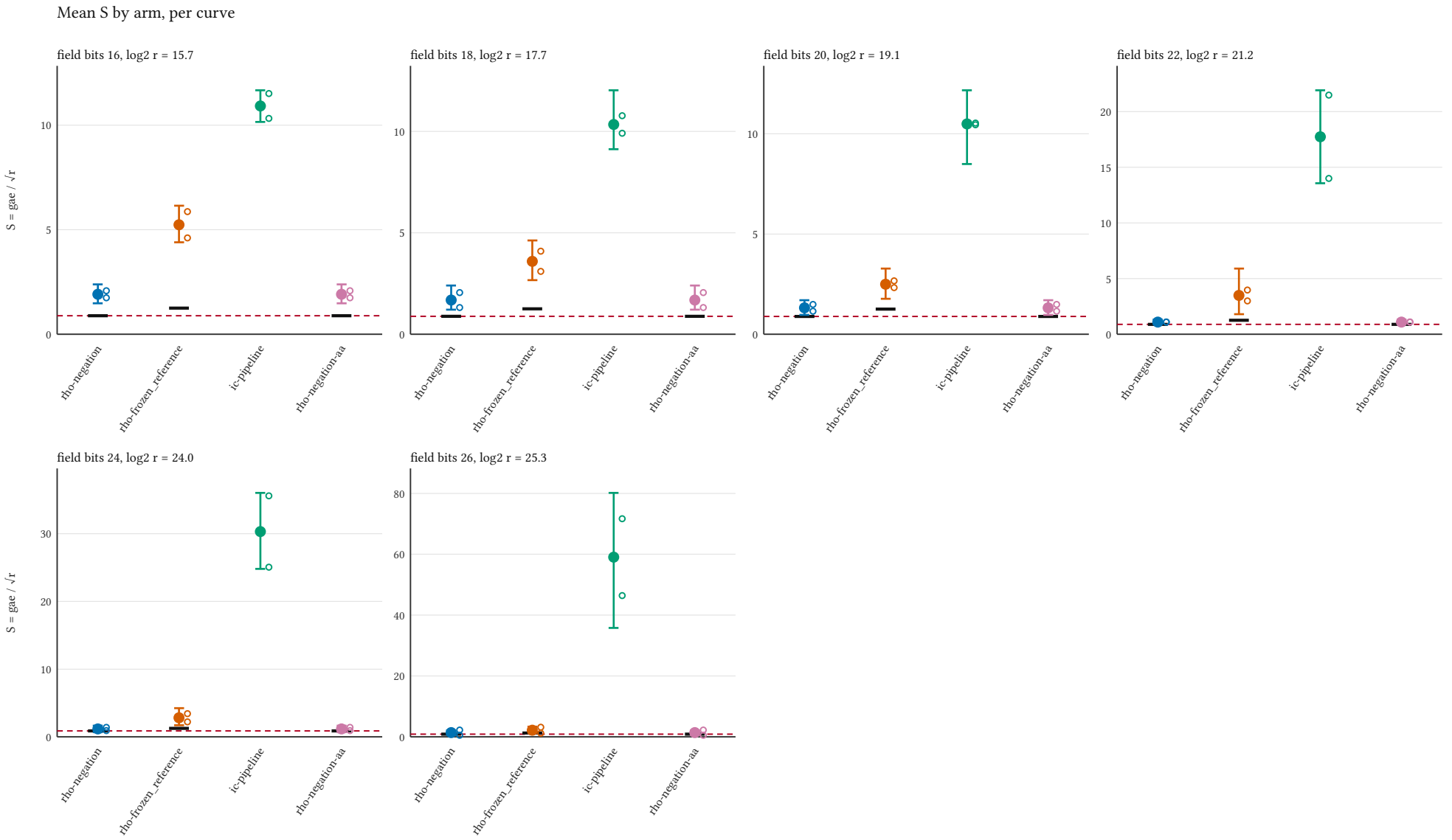
workload	curve	family	log2 r	A	target index	target seed
Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	prime	25.346	2	0	20261002
Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	prime	25.346	2	1	20261002

Runs

Planned 192 (48 warm-up, 144 measured); records 192 (144 measured); verified 144 (144 with matches_target); not verified 0; planned without a record 0.

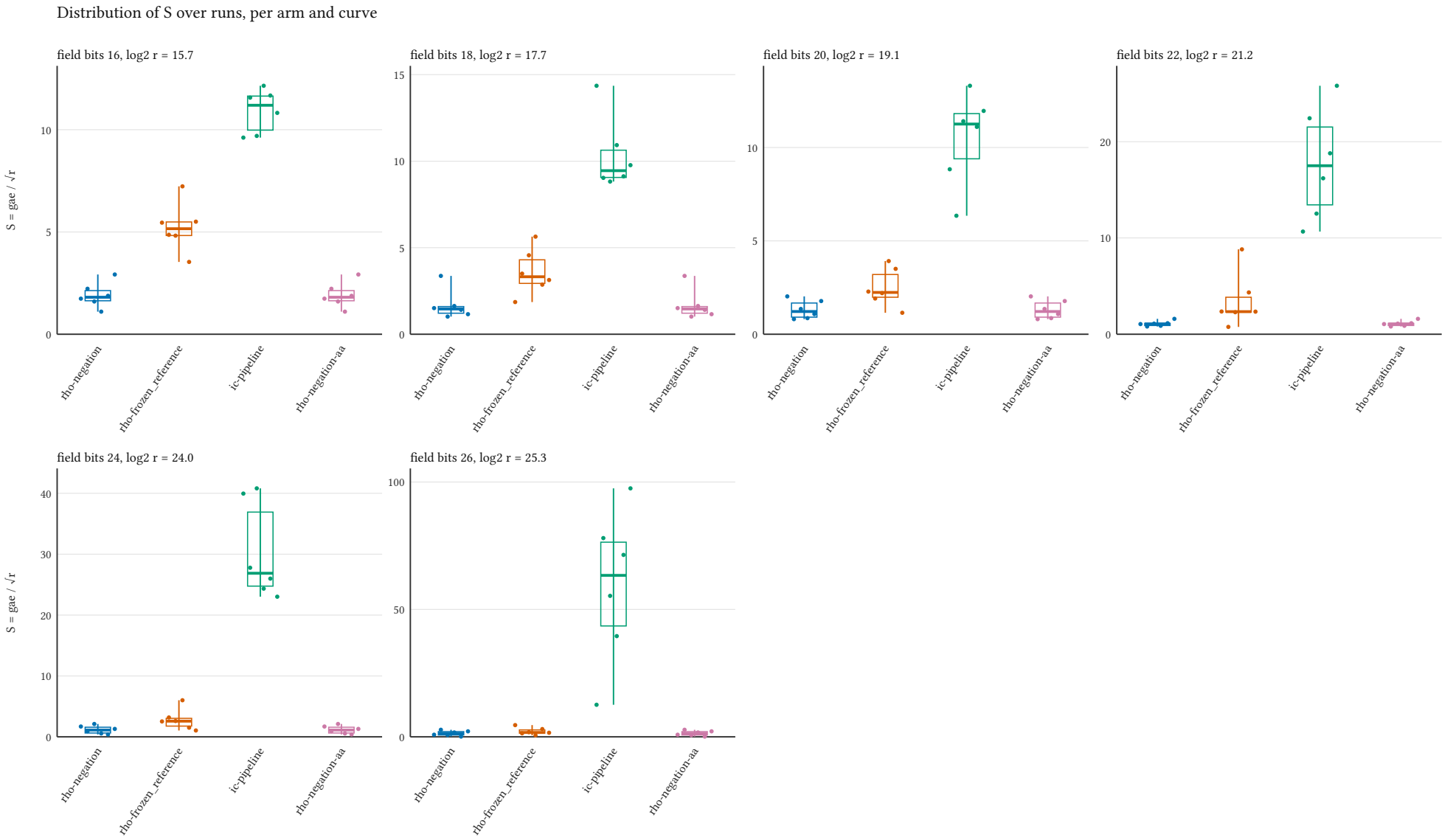
No record has a status other than verified.

Graphs



Session ECBS1hec2206c6baf4, n = 144 verified runs. Unit: S = gae / √r.

Figure 1: Mean S by arm per curve with 95% bootstrap whiskers, the floor and expected S; session ECBS1hec2206c6baf4, n = 144 verified runs, unit gae (S = gae / √r)



Session ECBS1hec2206c6baf4, n = 144 verified runs. Unit: S = gae / √r.

Figure 2: Distribution of S over the verified runs, per arm and curve; session ECBS1hec2206c6baf4, n = 144 verified runs, unit gae (S = gae / √r)

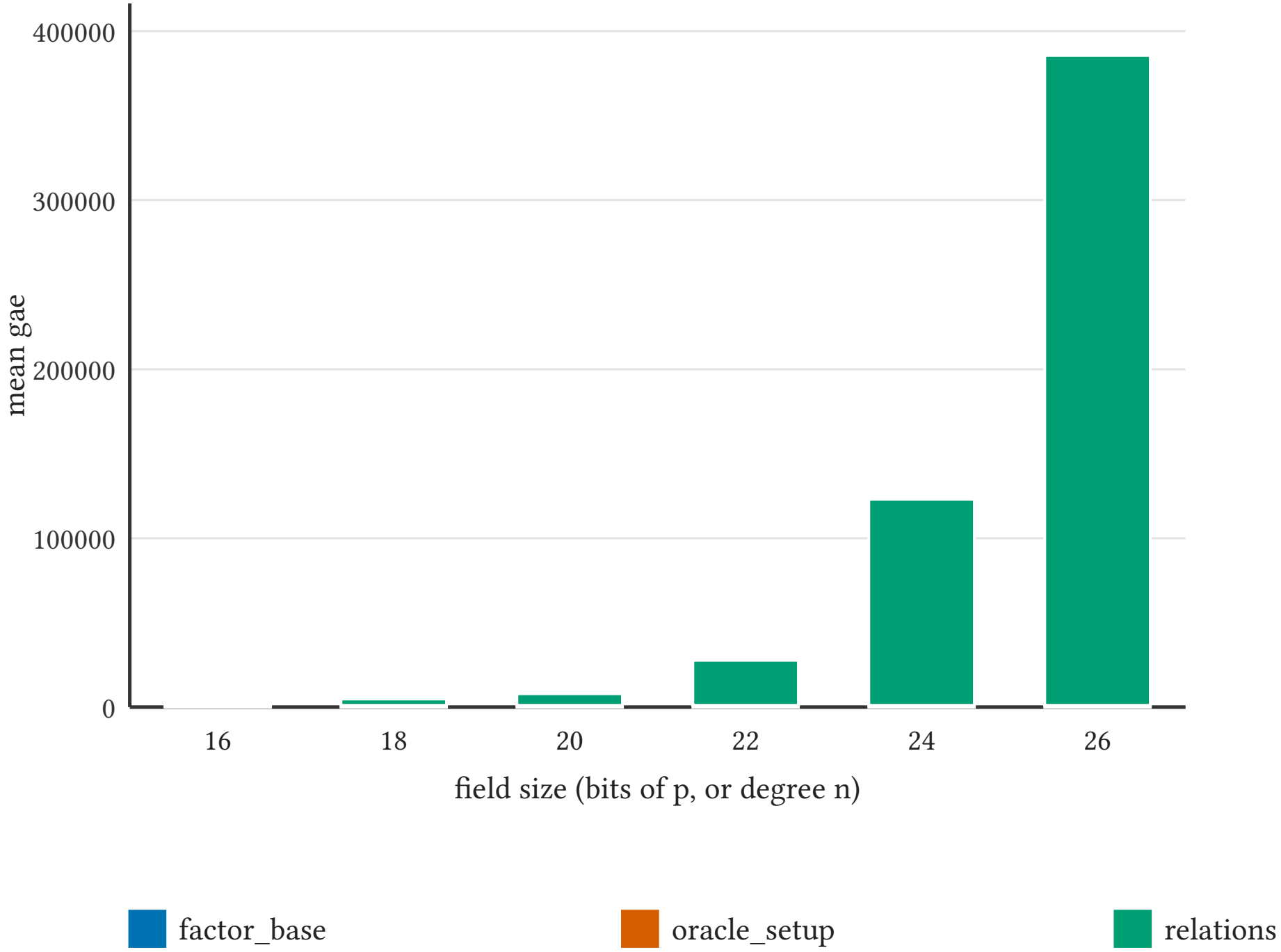
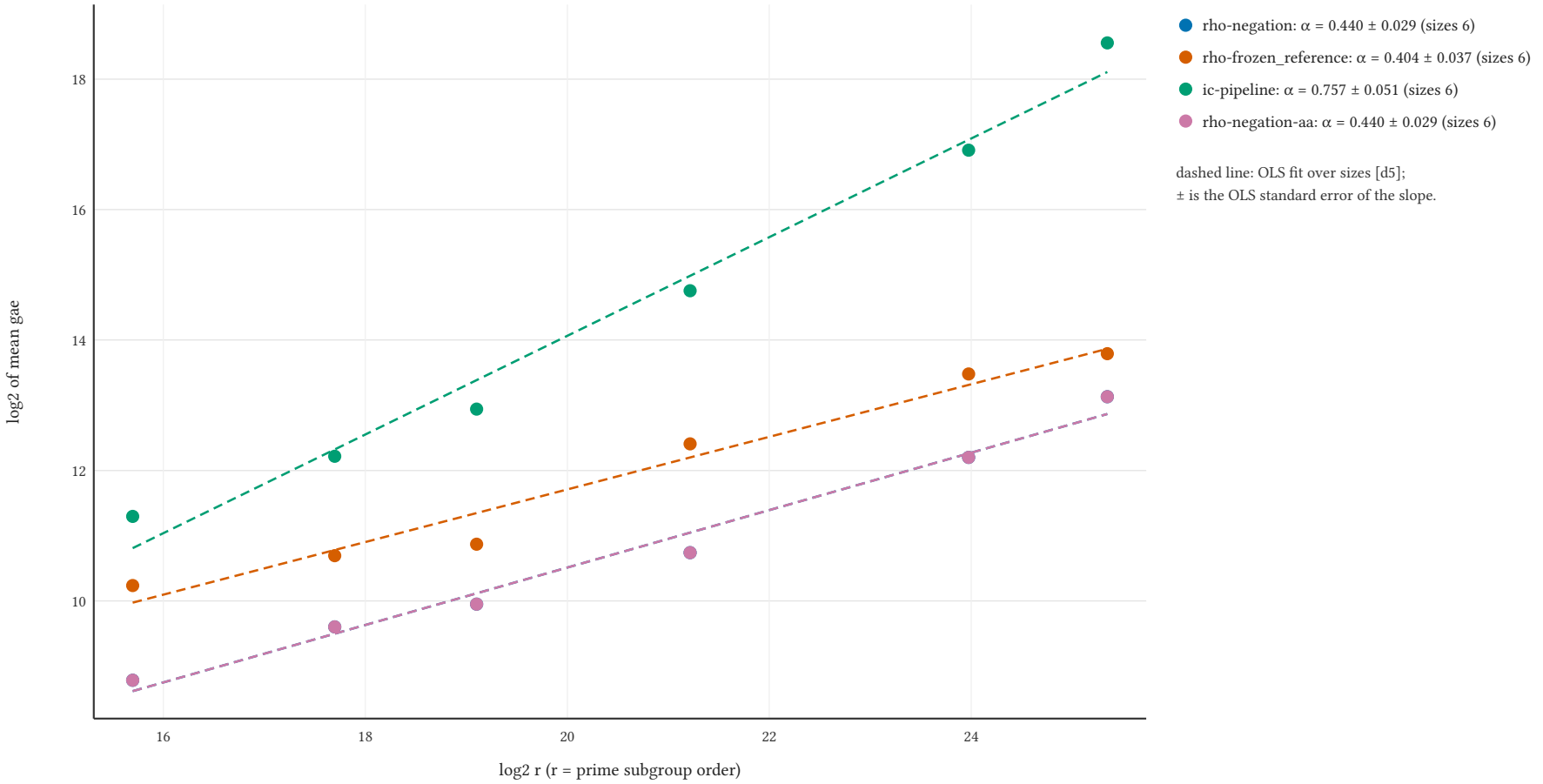


Figure 3: Mean gae by phase, index-calculus arms; session ECBS1hec2206c6baf4, n = 144 verified runs, unit gae (S = gae / √r)



Session ECBS1hec2206c6baf4, n = 144 verified runs over 6 sizes. Unit: gae (counted group operations).

Figure 4: log2 mean gae against log2 r with OLS fits and fitted exponents; session ECBS1hec2206c6baf4, n = 144 verified runs, unit gae (S = gae / √r)

Results per arm and curve

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-negation	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.9113	1.8093	1.1021	2.9201	0.6153	[1.4789, 2.3879]	0.8862	0.8862
rho-negation	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.6851	1.4612	1.0219	3.3738	0.8574	[1.2121, 2.4036]	0.8862	0.8862
rho-negation	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.3185	1.2167	0.8081	2.0295	0.4981	[0.9630, 1.6992]	0.8862	0.8862
rho-negation	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.0958	1.0726	0.8048	1.5993	0.2796	[0.9151, 1.3196]	0.8862	0.8862
rho-negation	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.1585	1.1079	0.3845	2.1159	0.6693	[0.6807, 1.6464]	0.8862	0.8862

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-negation	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.3732	1.2605	0.0859	2.7800	1.0178	[0.6304, 2.1375]	0.8862	0.8862
rho-frozen_reference	icv1-fp16-t295-8d3c3165	15.697	6 / 6	5.2349	5.1612	3.5362	7.2287	1.2076	[4.3982, 6.1454]	0.8862	1.2533
rho-frozen_reference	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	3.5951	3.3195	1.8572	5.6410	1.3338	[2.6647, 4.6264]	0.8862	1.2533
rho-frozen_reference	icv1-fp20-t727-cd198a38	19.101	6 / 6	2.4933	2.2421	1.1467	3.9149	1.0317	[1.7666, 3.2747]	0.8862	1.2533
rho-frozen_reference	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	3.4853	2.3512	0.7638	8.8184	2.8485	[1.8021, 5.9044]	0.8862	1.2533
rho-frozen_reference	icv1-fp24-t1059-6df599df	23.974	6 / 6	2.8141	2.5693	1.0392	6.0163	1.7543	[1.7059, 4.2442]	0.8862	1.2533
rho-frozen_reference	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	2.1687	1.7098	0.6005	4.5750	1.4184	[1.2305, 3.3244]	0.8862	1.2533
ic-pipeline	icv1-fp16-t295-8d3c3165	15.697	6 / 6	10.9189	11.1945	9.6151	12.1490	1.0683	[10.1546, 11.6703]	0.8862	–
ic-pipeline	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	10.3422	9.4509	8.8282	14.3608	2.1110	[9.1207, 12.0291]	0.8862	–
ic-pipeline	icv1-fp20-t727-cd198a38	19.101	6 / 6	10.4922	11.2547	6.3471	13.3075	2.4984	[8.4886, 12.1726]	0.8862	–
ic-pipeline	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	17.7379	17.4940	10.6650	25.8082	5.7934	[13.5629, 21.9129]	0.8862	–
ic-pipeline	icv1-fp24-t1059-6df599df	23.974	6 / 6	30.3175	26.8804	23.0283	40.8224	7.9656	[24.8057, 36.0301]	0.8862	–
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	59.0447	63.3354	12.5972	97.4978	30.1434	[35.7995, 80.1809]	0.8862	–
rho-negation-aa	icv1-fp16-t295-8d3c3165	15.697	6 / 6	1.9113	1.8093	1.1021	2.9201	0.6153	[1.4789, 2.3879]	0.8862	0.8862
rho-negation-aa	icv1-fp18-tm337-d28d5e09	17.697	6 / 6	1.6851	1.4612	1.0219	3.3738	0.8574	[1.2121, 2.4036]	0.8862	0.8862
rho-negation-aa	icv1-fp20-t727-cd198a38	19.101	6 / 6	1.3185	1.2167	0.8081	2.0295	0.4981	[0.9630, 1.6992]	0.8862	0.8862
rho-negation-aa	icv1-fp22-tm1385-475dcb5f	21.216	6 / 6	1.0958	1.0726	0.8048	1.5993	0.2796	[0.9151, 1.3196]	0.8862	0.8862
rho-negation-aa	icv1-fp24-t1059-6df599df	23.974	6 / 6	1.1585	1.1079	0.3845	2.1159	0.6693	[0.6807, 1.6464]	0.8862	0.8862
rho-negation-aa	icv1-fp26-tm1775-7e8fb6df	25.346	6 / 6	1.3732	1.2605	0.0859	2.7800	1.0178	[0.6304, 2.1375]	0.8862	0.8862

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation	icv1-fp16-t295-8d3c3165	6 / 6	440.5	2.1567	15.0	254.5	1161.0	533.8	440.5	5.5834	38.0620	informative, L1
rho-negation	icv1-fp18-tm337-d28d5e09	6 / 6	776.7	1.9014	33.8	545.3	1845.3	874.0	776.7	5.3042	37.0755	informative, L1
rho-negation	icv1-fp20-t727-cd198a38	6 / 6	988.8	1.4878	19.5	751.0	2275.7	1088.2	988.8	5.6120	38.4800	admitted (≥L2)
rho-negation	icv1-fp22-tm1385-475dcb5f	6 / 6	1710.2	1.2364	44.3	1384.0	3749.8	1820.0	1710.2	6.3785	38.8900	admitted (≥L2)
rho-negation	icv1-fp24-t1059-6df599df	6 / 6	4703.2	1.3072	60.7	4127.3	10066.3	4923.2	4703.2	6.6478	36.9135	admitted (≥L2)
rho-negation	icv1-fp26-tm1775-7e8fb6df	6 / 6	8968.2	1.5495	125.2	8193.8	18624.8	9197.7	8968.2	8.3358	44.7000	admitted (≥L2)
rho-frozen_reference	icv1-fp16-t295-8d3c3165	6 / 6	1206.5	5.9070	9.2	311.5	4168.0	1791.5	1206.5	5.4493	34.4450	informative, L1
rho-frozen_reference	icv1-fp18-tm337-d28d5e09	6 / 6	1657.0	4.0566	11.2	576.3	5409.0	2355.3	1657.0	5.0414	35.5680	admitted (≥L2)
rho-frozen_reference	icv1-fp20-t727-cd198a38	6 / 6	1869.8	2.8134	8.3	931.3	5582.2	2484.0	1869.8	5.1556	36.0065	informative, L1
rho-frozen_reference	icv1-fp22-tm1385-475dcb5f	6 / 6	5439.5	3.9327	8.0	4409.5	12879.5	6106.3	5439.5	6.8651	39.1080	admitted (≥L2)
rho-frozen_reference	icv1-fp24-t1059-6df599df	6 / 6	11424.8	3.1754	14.0	9952.3	25739.7	12388.2	11424.8	7.8578	42.8090	admitted (≥L2)
rho-frozen_reference	icv1-fp26-tm1775-7e8fb6df	6 / 6	14163.7	2.4471	11.5	12804.0	31019.8	15061.2	14163.7	8.7746	45.7905	admitted (≥L2)
ic-pipeline	icv1-fp16-t295-8d3c3165	6 / 6	2516.5	12.3207	–	–	–	–	–	0.7083	39.6220	admitted (≥L2)
ic-pipeline	icv1-fp18-tm337-d28d5e09	6 / 6	4766.8	11.6700	–	–	–	–	–	1.1821	42.3485	informative, L1
ic-pipeline	icv1-fp20-t727-cd198a38	6 / 6	7868.7	11.8392	–	–	–	–	–	1.8137	42.9085	informative, L1
ic-pipeline	icv1-fp22-tm1385-475dcb5f	6 / 6	27683.7	20.0151	–	–	–	–	–	6.2468	57.4190	admitted (≥L2)
ic-pipeline	icv1-fp24-t1059-6df599df	6 / 6	123084.0	34.2096	–	–	–	–	–	28.4641	81.5785	informative, L1
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	6 / 6	385620.7	66.6248	–	–	–	–	–	137.3849	189.2580	informative, L1
rho-negation-aa	icv1-fp16-t295-8d3c3165	6 / 6	440.5	2.1567	15.0	254.5	1161.0	533.8	440.5	5.3031	34.7440	admitted (≥L2)
rho-negation-aa	icv1-fp18-tm337-d28d5e09	6 / 6	776.7	1.9014	33.8	545.3	1845.3	874.0	776.7	5.6043	41.8080	admitted (≥L2)
rho-negation-aa	icv1-fp20-t727-cd198a38	6 / 6	988.8	1.4878	19.5	751.0	2275.7	1088.2	988.8	5.1586	36.9715	informative, L1
rho-negation-aa	icv1-fp22-tm1385-475dcb5f	6 / 6	1710.2	1.2364	44.3	1384.0	3749.8	1820.0	1710.2	6.3930	39.6375	admitted (≥L2)
rho-negation-aa	icv1-fp24-t1059-6df599df	6 / 6	4703.2	1.3072	60.7	4127.3	10066.3	4923.2	4703.2	7.6107	39.5200	admitted (≥L2)
rho-negation-aa	icv1-fp26-tm1775-7e8fb6df	6 / 6	8968.2	1.5495	125.2	8193.8	18624.8	9197.7	8968.2	8.7888	44.8895	admitted (≥L2)

Results per arm and workload

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-negation	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.7414	1.7399	1.6054	1.8788	0.1367	[1.6054, 1.8788]	0.8862	0.8862
rho-negation	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	2.0812	2.2215	1.1021	2.9201	0.9171	[1.1021, 2.9201]	0.8862	0.8862
rho-negation	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.3148	1.4146	1.0219	1.5079	0.2579	[1.0219, 1.5079]	0.8862	0.8862
rho-negation	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	2.0554	1.6337	1.1586	3.3738	1.1662	[1.1586, 3.3738]	0.8862	0.8862
rho-negation	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.4877	1.3548	1.0787	2.0295	0.4891	[1.0787, 2.0295]	0.8862	0.8862
rho-negation	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.1494	0.8641	0.8081	1.7761	0.5435	[0.8081, 1.7761]	0.8862	0.8862
rho-negation	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0978	1.0944	1.0508	1.1482	0.0488	[1.0508, 1.1482]	0.8862	0.8862
rho-negation	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0937	0.8772	0.8048	1.5993	0.4393	[0.8048, 1.5993]	0.8862	0.8862
rho-negation	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.3951	1.6850	0.3845	2.1159	0.9014	[0.3845, 2.1159]	0.8862	0.8862
rho-negation	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	0.9218	0.9197	0.5495	1.2961	0.3733	[0.5495, 1.2961]	0.8862	0.8862
rho-negation	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.5281	0.6728	0.0859	0.8258	0.3906	[0.0859, 0.8258]	0.8862	0.8862
rho-negation	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	2.2182	2.1793	1.6953	2.7800	0.5434	[1.6953, 2.7800]	0.8862	0.8862
rho-frozen_reference	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	4.6051	4.8206	3.5362	5.4584	0.9790	[3.5362, 5.4584]	0.8862	1.2533
rho-frozen_reference	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	5.8648	5.5018	4.8639	7.2287	1.2234	[4.8639, 7.2287]	0.8862	1.2533
rho-frozen_reference	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	3.0968	2.8682	1.8572	4.5649	1.3682	[1.8572, 4.5649]	0.8862	1.2533
rho-frozen_reference	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	4.0934	3.5039	3.1351	5.6410	1.3529	[3.1351, 5.6410]	0.8862	1.2533
rho-frozen_reference	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	2.6624	2.2841	2.2001	3.5029	0.7291	[2.2001, 3.5029]	0.8862	1.2533
rho-frozen_reference	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	2.3242	1.9108	1.1467	3.9149	1.4296	[1.1467, 3.9149]	0.8862	1.2533
rho-frozen_reference	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	2.9952	2.3585	2.2913	4.3359	1.1615	[2.2913, 4.3359]	0.8862	1.2533
rho-frozen_reference	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	3.9753	2.3438	0.7638	8.8184	4.2680	[0.7638, 8.8184]	0.8862	1.2533
rho-frozen_reference	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	2.2136	2.5338	1.5020	2.6048	0.6172	[1.5020, 2.6048]	0.8862	1.2533
rho-frozen_reference	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	3.4147	3.1886	1.0392	6.0163	2.4962	[1.0392, 6.0163]	0.8862	1.2533
rho-frozen_reference	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	3.1383	3.0357	1.8043	4.5750	1.3882	[1.8043, 4.5750]	0.8862	1.2533
rho-frozen_reference	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	1.1991	1.3813	0.6005	1.6154	0.5314	[0.6005, 1.6154]	0.8862	1.2533
ic-pipeline	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	10.3252	9.6932	9.6151	11.6674	1.1630	[9.6151, 11.6674]	0.8862	–
ic-pipeline	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	11.5126	11.5676	10.8213	12.1490	0.6656	[10.8213, 12.1490]	0.8862	–
ic-pipeline	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	10.7729	9.1298	8.8282	14.3608	3.1108	[8.8282, 14.3608]	0.8862	–
ic-pipeline	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	9.9116	9.7720	9.0365	10.9262	0.9526	[9.0365, 10.9262]	0.8862	–
ic-pipeline	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	10.4478	11.1007	8.8339	11.4087	1.4061	[8.8339, 11.4087]	0.8862	–
ic-pipeline	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	10.5367	11.9554	6.3471	13.3075	3.6907	[6.3471, 13.3075]	0.8862	–
ic-pipeline	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	13.9934	12.5296	10.6650	18.7857	4.2536	[10.6650, 18.7857]	0.8862	–
ic-pipeline	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	21.4823	22.4366	16.2023	25.8082	4.8735	[16.2023, 25.8082]	0.8862	–
ic-pipeline	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	35.5879	39.9526	25.9887	40.8224	8.3245	[25.9887, 40.8224]	0.8862	–
ic-pipeline	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	25.0471	24.3409	23.0283	27.7721	2.4495	[23.0283, 27.7721]	0.8862	–
ic-pipeline	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	46.4227	55.2971	12.5972	71.3738	30.3766	[12.5972, 71.3738]	0.8862	–
ic-pipeline	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	71.6667	77.9979	39.5044	97.4978	29.5106	[39.5044, 97.4978]	0.8862	–
rho-negation-aa	W43a0abf0d808	icv1-fp16-t295-8d3c3165	3 / 3	1.7414	1.7399	1.6054	1.8788	0.1367	[1.6054, 1.8788]	0.8862	0.8862
rho-negation-aa	Wd0d7d5ab0c1f	icv1-fp16-t295-8d3c3165	3 / 3	2.0812	2.2215	1.1021	2.9201	0.9171	[1.1021, 2.9201]	0.8862	0.8862
rho-negation-aa	Wbfca0f4fc222	icv1-fp18-tm337-d28d5e09	3 / 3	1.3148	1.4146	1.0219	1.5079	0.2579	[1.0219, 1.5079]	0.8862	0.8862
rho-negation-aa	Wd5517e0db0bc	icv1-fp18-tm337-d28d5e09	3 / 3	2.0554	1.6337	1.1586	3.3738	1.1662	[1.1586, 3.3738]	0.8862	0.8862
rho-negation-aa	W141767324d6e	icv1-fp20-t727-cd198a38	3 / 3	1.4877	1.3548	1.0787	2.0295	0.4891	[1.0787, 2.0295]	0.8862	0.8862
rho-negation-aa	W8d7a772f8ccd	icv1-fp20-t727-cd198a38	3 / 3	1.1494	0.8641	0.8081	1.7761	0.5435	[0.8081, 1.7761]	0.8862	0.8862
rho-negation-aa	W9a300e433cac	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0978	1.0944	1.0508	1.1482	0.0488	[1.0508, 1.1482]	0.8862	0.8862
rho-negation-aa	W6e13d2547daa	icv1-fp22-tm1385-475dcb5f	3 / 3	1.0937	0.8772	0.8048	1.5993	0.4393	[0.8048, 1.5993]	0.8862	0.8862
rho-negation-aa	Wec44ca6ca670	icv1-fp24-t1059-6df599df	3 / 3	1.3951	1.6850	0.3845	2.1159	0.9014	[0.3845, 2.1159]	0.8862	0.8862

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-negation-aa	W3256bf1f8f74	icv1-fp24-t1059-6df599df	3 / 3	0.9218	0.9197	0.5495	1.2961	0.3733	[0.5495, 1.2961]	0.8862	0.8862
rho-negation-aa	Wf9c5da9613ce	icv1-fp26-tm1775-7e8fb6df	3 / 3	0.5281	0.6728	0.0859	0.8258	0.3906	[0.0859, 0.8258]	0.8862	0.8862
rho-negation-aa	Wbee2c4f33e98	icv1-fp26-tm1775-7e8fb6df	3 / 3	2.2182	2.1793	1.6953	2.7800	0.5434	[1.6953, 2.7800]	0.8862	0.8862

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation	W43a0abf0d808	3 / 3	401.3	1.9649	14.3	225.7	1075.7	492.3	401.3	5.3980	46.9980	informative, L1
rho-negation	Wd0d7d5ab0c1f	3 / 3	479.7	2.3484	15.7	283.3	1246.3	575.3	479.7	5.7196	35.5650	admitted (≥L2)
rho-negation	Wbfca0f4fc222	3 / 3	606.0	1.4836	23.3	400.7	1494.0	700.0	606.0	5.4198	34.4450	informative, L1
rho-negation	Wd5517e0db0bc	3 / 3	947.3	2.3192	44.3	690.0	2196.7	1048.0	947.3	5.1886	38.7150	admitted (≥L2)
rho-negation	W141767324d6e	3 / 3	1115.7	1.6786	21.0	862.0	2536.3	1217.3	1115.7	5.3216	38.7690	admitted (≥L2)
rho-negation	W8d7a772f8ccd	3 / 3	862.0	1.2970	18.0	640.0	2015.0	959.0	862.0	5.8869	38.1910	admitted (≥L2)
rho-negation	W9a300e433cac	3 / 3	1713.3	1.2387	46.0	1392.7	3761.7	1825.0	1713.3	6.5812	42.1850	admitted (≥L2)
rho-negation	W6e13d2547daa	3 / 3	1707.0	1.2341	42.7	1375.3	3738.0	1815.0	1707.0	6.1757	35.5950	admitted (≥L2)
rho-negation	Wec44ca6ca670	3 / 3	5664.0	1.5742	78.0	5036.7	11991.0	5885.0	5664.0	6.9309	40.6820	admitted (≥L2)
rho-negation	W3256bf1f8f74	3 / 3	3742.3	1.0401	43.3	3218.0	8141.7	3961.3	3742.3	6.3648	36.0050	admitted (≥L2)
rho-negation	Wf9c5da9613ce	3 / 3	3449.3	0.5960	48.3	2925.3	7569.7	3673.0	3449.3	7.9729	39.7410	admitted (≥L2)
rho-negation	Wbee2c4f33e98	3 / 3	14487.0	2.5030	202.0	13462.3	29680.0	14722.3	14487.0	8.4602	49.6590	admitted (≥L2)
rho-frozen_reference	W43a0abf0d808	3 / 3	1061.3	5.1963	8.0	212.3	3779.7	1613.7	1061.3	5.5930	34.2380	informative, L1
rho-frozen_reference	Wd0d7d5ab0c1f	3 / 3	1351.7	6.6177	10.3	410.7	4556.3	1969.3	1351.7	5.3056	34.6520	admitted (≥L2)
rho-frozen_reference	Wbfca0f4fc222	3 / 3	1427.3	3.4943	9.7	429.3	4794.7	2074.0	1427.3	4.9819	35.8130	admitted (≥L2)
rho-frozen_reference	Wd5517e0db0bc	3 / 3	1886.7	4.6189	12.7	723.3	6023.3	2636.7	1886.7	5.1009	35.3230	admitted (≥L2)
rho-frozen_reference	W141767324d6e	3 / 3	1996.7	3.0042	8.7	1038.7	5890.3	2629.0	1996.7	5.1451	43.6850	admitted (≥L2)
rho-frozen_reference	W8d7a772f8ccd	3 / 3	1743.0	2.6225	8.0	824.0	5274.0	2339.0	1743.0	5.1662	33.2620	informative, L1
rho-frozen_reference	W9a300e433cac	3 / 3	4674.7	3.3797	7.7	3667.0	11298.3	5324.3	4674.7	6.8068	42.8020	admitted (≥L2)
rho-frozen_reference	W6e13d2547daa	3 / 3	6204.3	4.4857	8.3	5152.0	14460.7	6888.3	6204.3	6.9233	35.4140	admitted (≥L2)
rho-frozen_reference	Wec44ca6ca670	3 / 3	8986.7	2.4977	11.3	7700.3	20496.3	9827.7	8986.7	8.0254	48.8240	admitted (≥L2)
rho-frozen_reference	W3256bf1f8f74	3 / 3	13863.0	3.8530	16.7	12204.3	30983.0	14948.7	13863.0	7.6903	37.0850	admitted (≥L2)
rho-frozen_reference	Wf9c5da9613ce	3 / 3	20496.3	3.5412	15.3	18840.7	44266.7	21587.7	20496.3	10.2848	42.5110	admitted (≥L2)
rho-frozen_reference	Wbee2c4f33e98	3 / 3	7831.0	1.3530	7.7	6767.3	17773.0	8534.7	7831.0	7.9612	49.0700	admitted (≥L2)
ic-pipeline	W43a0abf0d808	3 / 3	2379.7	11.6508	–	–	–	–	–	0.6910	37.0280	admitted (≥L2)
ic-pipeline	Wd0d7d5ab0c1f	3 / 3	2653.3	12.9906	–	–	–	–	–	0.7729	42.2160	admitted (≥L2)
ic-pipeline	Wbfca0f4fc222	3 / 3	4965.3	12.1559	–	–	–	–	–	1.1602	39.9270	admitted (≥L2)
ic-pipeline	Wd5517e0db0bc	3 / 3	4568.3	11.1840	–	–	–	–	–	1.2039	44.3190	informative, L1
ic-pipeline	W141767324d6e	3 / 3	7835.3	11.7891	–	–	–	–	–	1.6483	44.2240	informative, L1
ic-pipeline	W8d7a772f8ccd	3 / 3	7902.0	11.8894	–	–	–	–	–	1.9790	41.5680	admitted (≥L2)
ic-pipeline	W9a300e433cac	3 / 3	21839.7	15.7899	–	–	–	–	–	4.7269	56.2820	admitted (≥L2)
ic-pipeline	W6e13d2547daa	3 / 3	33527.7	24.2402	–	–	–	–	–	7.9611	60.2380	admitted (≥L2)
ic-pipeline	Wec44ca6ca670	3 / 3	144481.0	40.1567	–	–	–	–	–	43.9691	85.3270	admitted (≥L2)
ic-pipeline	W3256bf1f8f74	3 / 3	101687.0	28.2626	–	–	–	–	–	25.6428	64.3390	informative, L1
ic-pipeline	Wf9c5da9613ce	3 / 3	303186.3	52.3824	–	–	–	–	–	119.4357	174.1510	informative, L1
ic-pipeline	Wbee2c4f33e98	3 / 3	468055.0	80.8672	–	–	–	–	–	184.1624	236.9760	admitted (≥L2)
rho-negation-aa	W43a0abf0d808	3 / 3	401.3	1.9649	14.3	225.7	1075.7	492.3	401.3	5.1958	33.9070	admitted (≥L2)
rho-negation-aa	Wd0d7d5ab0c1f	3 / 3	479.7	2.3484	15.7	283.3	1246.3	575.3	479.7	5.9087	35.5810	admitted (≥L2)
rho-negation-aa	Wbfca0f4fc222	3 / 3	606.0	1.4836	23.3	400.7	1494.0	700.0	606.0	5.9241	43.1320	admitted (≥L2)
rho-negation-aa	Wd5517e0db0bc	3 / 3	947.3	2.3192	44.3	690.0	2196.7	1048.0	947.3	4.9093	31.5840	admitted (≥L2)
rho-negation-aa	W141767324d6e	3 / 3	1115.7	1.6786	21.0	862.0	2536.3	1217.3	1115.7	5.3444	37.5140	admitted (≥L2)

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation-aa	W8d7a772f8ccd	3 / 3	862.0	1.2970	18.0	640.0	2015.0	959.0	862.0	4.9152	31.1680	informative, L1
rho-negation-aa	W9a300e433cac	3 / 3	1713.3	1.2387	46.0	1392.7	3761.7	1825.0	1713.3	6.6398	38.6940	admitted (≥L2)
rho-negation-aa	W6e13d2547daa	3 / 3	1707.0	1.2341	42.7	1375.3	3738.0	1815.0	1707.0	5.8336	41.7460	admitted (≥L2)
rho-negation-aa	Wec44ca6ca670	3 / 3	5664.0	1.5742	78.0	5036.7	11991.0	5885.0	5664.0	7.5727	37.7220	admitted (≥L2)
rho-negation-aa	W3256bf1f8f74	3 / 3	3742.3	1.0401	43.3	3218.0	8141.7	3961.3	3742.3	7.7319	41.3180	admitted (≥L2)
rho-negation-aa	Wf9c5da9613ce	3 / 3	3449.3	0.5960	48.3	2925.3	7569.7	3673.0	3449.3	7.2840	43.9630	admitted (≥L2)
rho-negation-aa	Wbee2c4f33e98	3 / 3	14487.0	2.5030	202.0	13462.3	29680.0	14722.3	14487.0	10.5667	51.8780	admitted (≥L2)

Phases of the index-calculus arms

arm	curve	phase	runs	mean gae	share [d8]
ic-pipeline	icv1-fp16-t295-8d3c3165	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp16-t295-8d3c3165	oracle_setup	6	1056.0	42.0%
ic-pipeline	icv1-fp16-t295-8d3c3165	relations	6	1441.0	57.3%
ic-pipeline	icv1-fp16-t295-8d3c3165	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp16-t295-8d3c3165	verify	6	19.5	0.8%
ic-pipeline	icv1-fp18-tm337-d28d5e09	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp18-tm337-d28d5e09	oracle_setup	6	1056.0	22.2%
ic-pipeline	icv1-fp18-tm337-d28d5e09	relations	6	3688.3	77.4%
ic-pipeline	icv1-fp18-tm337-d28d5e09	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp18-tm337-d28d5e09	verify	6	22.5	0.5%
ic-pipeline	icv1-fp20-t727-cd198a38	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp20-t727-cd198a38	oracle_setup	6	1056.0	13.4%
ic-pipeline	icv1-fp20-t727-cd198a38	relations	6	6787.7	86.3%
ic-pipeline	icv1-fp20-t727-cd198a38	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp20-t727-cd198a38	verify	6	25.0	0.3%
ic-pipeline	icv1-fp22-tm1385-475dcb5f	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp22-tm1385-475dcb5f	oracle_setup	6	1056.0	3.8%
ic-pipeline	icv1-fp22-tm1385-475dcb5f	relations	6	26598.7	96.1%
ic-pipeline	icv1-fp22-tm1385-475dcb5f	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp22-tm1385-475dcb5f	verify	6	29.0	0.1%
ic-pipeline	icv1-fp24-t1059-6df599df	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp24-t1059-6df599df	oracle_setup	6	1056.0	0.9%
ic-pipeline	icv1-fp24-t1059-6df599df	relations	6	121993.0	99.1%
ic-pipeline	icv1-fp24-t1059-6df599df	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp24-t1059-6df599df	verify	6	35.0	0.0%
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	factor_base	6	0.0	0.0%
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	oracle_setup	6	1056.0	0.3%
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	relations	6	384527.7	99.7%
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	linear_algebra	6	0.0	0.0%
ic-pipeline	icv1-fp26-tm1775-7e8fb6df	verify	6	37.0	0.0%

Scaling

arm	sizes	verified runs	α [d5]	se(α)	log2 C	R^2
rho-negation	6	36	0.440	0.029	1.707	0.9828
rho-frozen_reference	6	36	0.404	0.037	3.639	0.9670
ic-pipeline	6	36	0.757	0.051	−1.068	0.9820

arm	sizes	verified runs	α [d5]	se(α)	log2 C	R^2
rho-negation-aa	6	36	0.440	0.029	1.707	0.9828

Raw records

`research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/prime-ic/records.jsonl` (192 records; ids in `raw_records.csv` and `report.json`); child stderr in `research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/prime-ic/exec/` (0 file(s)).

Derivations

- **[d1]** S = cost.s of each verified, non-warm-up record (`total_gae / √r`, stored by the runner); mean, median, min, max and sample stdev (stats.rs) are over those records.
- **[d2]** 95% percentile bootstrap interval of the mean S: 4000 resamples of the verified runs of the cell with replacement (stats::bootstrap_ci, seed 20261002). A cell with one run has no interval. Not a significance test.
- **[d3]** total gae and S / floor are means over the verified runs of the cell of cost.total_gae and boundaries.ratio_to_floor; floor S is boundaries.floor_s = √(π / 2A), A = automorphisms the curve offers a generic method.
- **[d4]** expected S is methods::expected_s(method, A): the closed form the method's mean S approaches as r grows (set-up omitted). A reference mark, not a measurement.
- **[d5]** scaling: per arm, per curve, the mean gae over the arm's verified runs on that curve; the line is ordinary least squares of log2(mean gae) on log2(r) over the curves (bounds::ols), slope α, intercept log2 C so gae = C·r^α. The standard error of α is √(SS_res/(sizes−2)/S_xx) from those residuals (none below 3 sizes). Counted operations, one point per size: it is not the cluster-bootstrap interval of `ecbench bound fit`, which resamples runs and sizes.
- **[d6]** counters (distinguished points, steps) and field operations (muls, sqrs, invs) are means over the cell's verified runs of the counters the record carries (steps = `steps`, or baby + giant steps for BSGS); a dash means the method did not count it (unknown, not zero).
- **[d7]** wall and CPU are medians over the cell's verified runs of time.solve_wall_ns and time.user_ns + sys_ns. The figure is admitted only when every run behind it earned the level the spec requires (compare.rs's gate); otherwise it is informative at the lowest level earned, and never a result.
- **[d8]** phase share = the phase's mean gae over the sum of the arm's mean gae across phases on that curve, over verified runs.